

COMP534 Assignment 2: 3-Class Lung X-Ray Classification

Normal / Opacity / Pneumonia

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1 Experimental Protocol

The dataset contains 3,475 chest X-ray images (299×299 px) across three classes: Normal (1,250, 36.0%), Opacity (1,125, 32.4%), and Pneumonia (1,100, 31.7%), yielding a mild imbalance ratio of 1.14:1.

- **Data Partitioning:** A **stratified 80/20 hold-out split** (`random_state=42`) creates a training pool (2,780) and a locked test set (695), with stratification preserving class proportions to prevent near-zero minority representation in small partitions [1].
- **Validation Strategy:** On the 80% pool, **stratified 5-fold cross-validation** ($\sim 2,224$ train / ~ 556 val per fold) provides a low-variance generalisation estimate—essential for only 3,475 images where a single random split would yield high variance.
- **Evaluation Metric:** Model selection uses **mean validation macro-F1** across folds rather than accuracy, which misleads under class imbalance by being dominated by the plurality class [2].
- **Final Evaluation:** The best *configuration* (not weights) is retrained on the full 80% pool and evaluated *once* on the locked test set, following the strict separation advocated by Hastie et al. [3] to prevent optimistic bias from repeated test-set access.

All random number generators are seeded at 42 with `cuda.deterministic=True` for full reproducibility.

2 Data Loading, Preprocessing, and Augmentation

2.1 Deterministic Preprocessing (All Splits)

Each image undergoes the following deterministic pipeline:

- **Resize:** Images are scaled to each model’s native resolution (300×300 for EfficientNet-B3, 256×256 for Swin V2, and 224×224 for MaxViT).
- **CLAHE [5]:** Applied with `clip_limit=2.0` and 8×8 tiles to enhance local contrast. This process reveals subtle pulmonary opacities often masked by the wide global dynamic range of X-ray imaging, a technique recently validated by EfficientNet achieving a macro-AUC of 0.906 on NIH ChestX-ray14 [6].
- **Normalize:** Pixel values are adjusted using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$), which is mandatory for utilizing pretrained weights.

Note: Critically, CLAHE is performed on `uint8` [0–255] intensities *before* the Normalize step shifts the distribution.

2.2 Stochastic Augmentation (Training Only)

Five augmentations model clinically realistic variability (Figure 1):

- **HorizontalFlip**($p=0.5$) exploits bilateral lung symmetry—a mirrored CXR is clinically indistinguishable from a real one (vertical flip is prohibited as anatomically non-physiological);
- **Rotate**($\pm 10^\circ$, $p=0.5$) models patient-positioning variance during acquisition [21];
- **RandomBrightnessContrast**(± 0.2 , $p=0.5$) simulates X-ray exposure variability across machines;
- **ElasticTransform**($\alpha = 1$, $\sigma = 50$, $p = 0.3$) models soft-tissue deformation and breathing-phase variation [20];

- **GaussNoise**($\text{var} \in [10, 50]$, $p = 0.3$) simulates quantum noise from low-dose acquisitions.

At batch level, **MixUp** [13] ($\alpha = 0.2$) and **CutMix** [14] ($\alpha = 1.0$, validated as optimal for medical images [15]) alternate randomly ($p = 0.5$). On mixed batches, label smoothing is disabled (MixUp already produces soft labels; combining over-regularises), while unmixed batches use $\varepsilon = 0.1$ [16].

2.3 Class Imbalance Handling

WeightedRandomSampler (WRS) assigns each sample weight $w_i = 1/\text{count}(\text{class}_i)$, ensuring approximately equal class representation per batch. Unlike class-weighted loss—which rescales gradient magnitudes but leaves AdamW’s moment-estimate covariance dominated by majority-class features—WRS balances gradient covariance in both magnitude *and* sample diversity simultaneously [17]. WRS is rebuilt per fold from fold-local labels to prevent data leakage. Loss weights are deliberately omitted from `CrossEntropyLoss` to avoid double-correction.

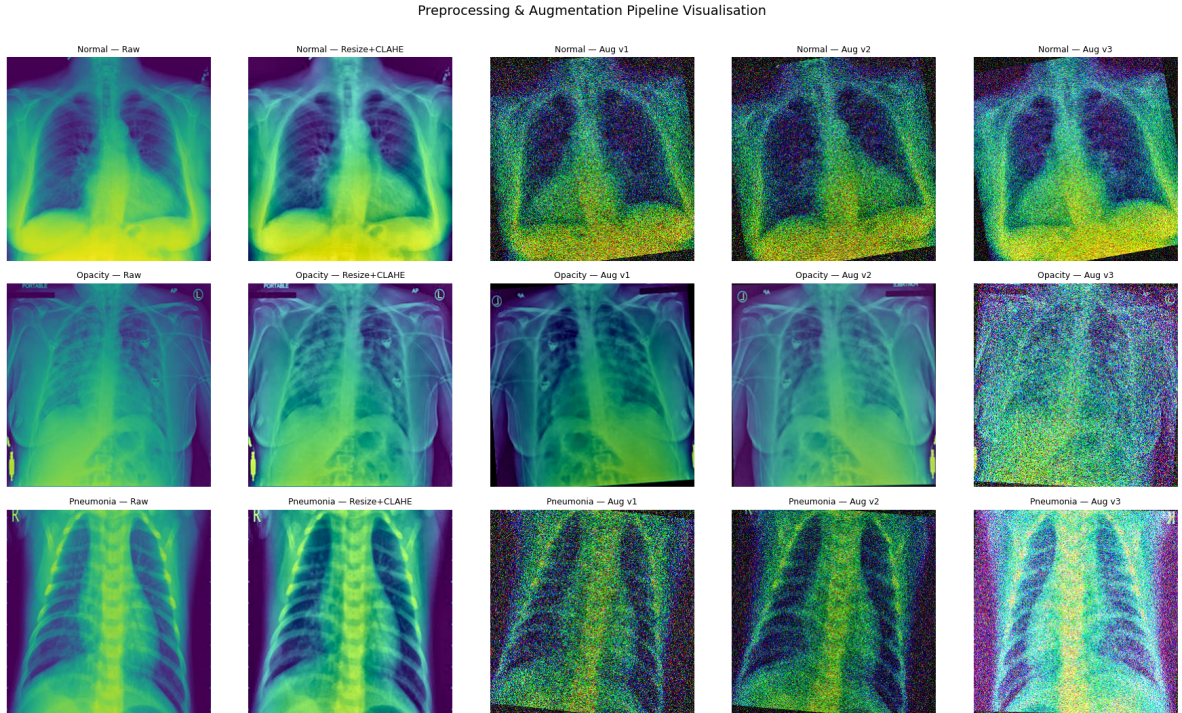


Figure 1: Preprocessing pipeline: Raw $\rightarrow$ Resize+CLAHE $\rightarrow$ stochastic augmentation variants per class.

3 Network Architectures, Strategies, and Training

3.1 Custom CNN: LightXRayCNN ($\sim 0.12\text{M}$ parameters)

The core design utilizes the following architectural components:

- **Depthwise-separable convolutions** [9]: A standard 3×3 conv ($C_{\text{in}} \times C_{\text{out}} \times 9$ params) is factored into a depthwise spatial filter ($C_{\text{in}} \times 9$ params) followed by a 1×1 pointwise channel-mixing filter ($C_{\text{in}} \times C_{\text{out}}$), yielding $\sim 8.9\times$ parameter reduction.
- **Architecture**: Consists of a stride-2 stem ($3 \rightarrow 32$ channels) and four DWSep blocks progressively expanding to 256 channels with MaxPool downsampling.
- **Global Average Pooling** (GAP): Replaces fully-connected layers for massive parameter reduction and implicit spatial regularisation [11].
- **Classification Head**: Includes Dropout(0.5) + Linear(256, 3).
- **GELU** [10]: Replaces ReLU for smoother gradient flow.

The design is deliberately parameter-efficient ($\sim 117\text{K}$ parameters), demonstrating that a lightweight architecture can achieve competitive performance ($\text{F1} = 0.8694$) when paired with strong preprocessing.

3.2 Existing Architectures (3 models $\times$ 3 strategies)

Model Architectures:

- **EfficientNet-B3** [4] (10.7M params, 300×300) applies compound scaling—jointly increasing depth, width, and resolution—more principled than scaling one dimension alone.
- **Swin Transformer V2 Small** [7] (49M params, 256×256) uses shifted-window self-attention with $O(n)$ complexity, producing hierarchical feature maps that capture long-range dependencies critical for distinguishing diffuse opacity from focal pneumonia consolidation.
- **MaxViT Tiny** [8] (31M params, 224×224) alternates blocked-local and dilated-global multi-axis attention, combining CNN-like locality with transformer-level global context.

Training Strategies:

- (1) **From scratch** (random init, uniform LR= 10^{-3});
- (2) **Feature extraction** (ImageNet backbone fully frozen, only classifier head trains at LR= 10^{-3});
- (3) **Fine-tuning** (early layers frozen to preserve low-level features, upper layers trained with differential LR—backbone at 10^{-4} , head at 10^{-3} [12]—preventing catastrophic forgetting while enabling domain adaptation).

3.3 Training Components

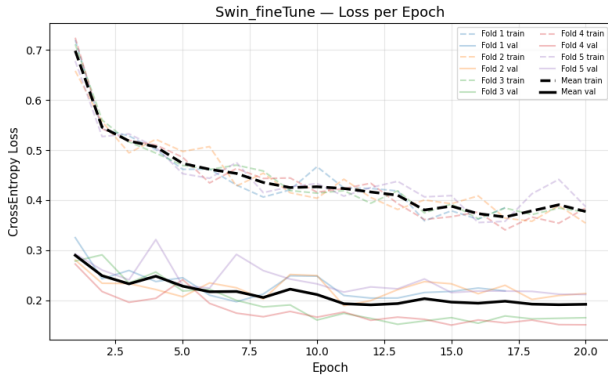
The training pipeline is configured with the following components:

- **Loss:** CrossEntropyLoss with conditional label smoothing ($\varepsilon = 0.1$ on unmixed batches), which softens hard targets to prevent overconfident predictions on a small dataset [16].
- **Optimiser:** AdamW [18] (weight decay 10^{-4}), which correctly decouples L_2 regularisation from the adaptive learning rate—unlike vanilla Adam where weight decay is entangled with gradient moments.
- **Gradient clipping** ($\|\nabla\|_{\max} = 1.0$) [19]: Stabilises fine-tuning where frozen/unfrozen layer boundaries create gradient scale mismatches.
- **Scheduler:** CosineAnnealingLR ($T_{\max} = 40$, $\eta_{\min} = 10^{-6}$)—a single cosine descent without restarts, since restarts spike the learning rate and trigger premature early stopping.
- **Early stopping** (patience=10 on val loss): Prevents overfitting on $\sim 2,224$ -image folds while restoring the best checkpoint.

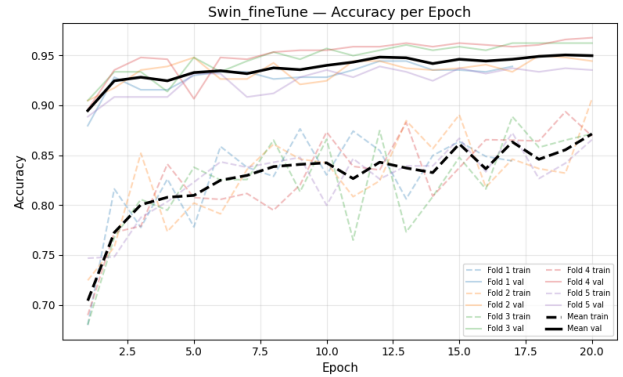
4 Cross-Validation Results and Model Comparison

Table 1: 5-fold CV results (mean $\pm$ std macro-F1). Best model selected by highest mean F1.

Configuration	Mean Val F1	Std	Per-fold F1
LightXRayCNN (scratch)	0.8694	0.0089	.865 .866 .877 .882 .857
EfficientNet-B3 (scratch)	0.9026	0.0092	.904 .903 .918 .891 .897
EfficientNet-B3 (feat. ext.)	0.8856	0.0044	.879 .891 .882 .888 .889
EfficientNet-B3 (fine-tune)	0.9379	0.0064	.931 .940 .943 .945 .929
Swin V2-S (scratch)	0.2924	0.0979	.338 .176 .358 .176 .413
Swin V2-S (feat. ext.)	0.8895	0.0084	.904 .892 .879 .884 .890
Swin V2-S (fine-tune)	0.9534	0.0106	.945 .951 .963 .968 .940
MaxViT-T (scratch)	0.6056	0.2328	.651 .860 .846 .350 .321
MaxViT-T (feat. ext.)	0.8865	0.0048	.896 .886 .885 .884 .881
MaxViT-T (fine-tune)	0.9431	0.0095	.938 .951 .954 .945 .927



(a) Loss per epoch



(b) Accuracy per epoch

Figure 2: Training curves for the best model (Swin V2-S fine-tune) across 5 folds (mean in black).

Three clear patterns emerge across strategies:

- **Fine-tuning dominates:** all three architectures achieve their best F1 under fine-tuning (Swin: 0.9534, MaxViT: 0.9431, EfficientNet: 0.9379), confirming that differential learning rates effectively adapt pretrained representations to the medical domain.
- **Scratch training fails for transformers:** Swin (0.2924) and MaxViT (0.6056) collapse without pretrained initialisation—their 49M and 31M parameters vastly exceed what 3,475 images can constrain, while EfficientNet’s CNN inductive biases (translation equivariance, local receptive fields) enable reasonable scratch performance (0.9026).
- **Feature extraction provides a stable floor** (~ 0.886 – 0.890) across all architectures with minimal variance, since frozen backbones cannot overfit.

The best model, **Swin V2-S fine-tune**, outperforms all others because shifted-window self-attention captures global spatial dependencies—critical for distinguishing diffuse opacity from focal consolidation—while fine-tuning adapts these representations to the CXR domain. The custom LightXRayCNN achieves 0.8694 with only 0.12M parameters, demonstrating the efficiency of depthwise-separable design.

5 Final Test-Set Evaluation

The best model was retrained on 90% of the training pool (10% internal holdout for early stopping) and evaluated *once* on the locked test set using 5-pass **TTA** [22] (original, H-flip, ± 5 rotation, brightness shift), averaging softmax probabilities before argmax.

We report five metrics, each capturing a genuinely different dimension of performance:

Table 2: Test-set metrics. Each captures a failure mode the others are blind to.

Metric	Value	What it uniquely measures
Macro-F1	0.9479	Precision-recall balance (hard predictions)
Cohen’s κ	0.9200	Agreement above chance (marginal baseline) [2]
MCC	0.9200	Symmetric confusion-matrix correlation (geometric baseline) [23]
Log Loss	0.1888	<i>Only</i> metric on probabilities; penalises overconfident errors logarithmically
Balanced Acc	0.9476	Pure macro recall; per-class detection equity

Table 3: Per-class precision, recall, and F1 on the test set.

Class	Prec.	Recall	F1	Support
Normal	0.9209	0.9320	0.9264	250
Opacity	0.9327	0.9244	0.9286	225
Pneumonia	0.9909	0.9864	0.9886	220
Macro avg	0.9482	0.9476	0.9479	695

The confusion matrix (Figure 3) reveals that Normal–Opacity confusion dominates errors (15 Normal→Opacity, 17 Opacity→Normal), reflecting their genuine clinical similarity as overlapping radiographic presentations. Critically, only **3 of 220 Pneumonia cases (1.4%)** were misclassified as Normal—the highest-severity error, representing a patient requiring treatment being incorrectly discharged. Pneumonia achieves near-perfect F1 (0.9886), indicating the model reliably detects the most clinically urgent pathology. **HiResCAM** [24] (Figure 4), chosen over standard GradCAM because Swin’s window-partitioned features require element-wise gradient $\times$ activation to preserve spatial resolution, confirms that activations concentrate on lung parenchyma: diffuse patterns for Opacity, focal consolidation for Pneumonia, and weak distributed activation for Normal.

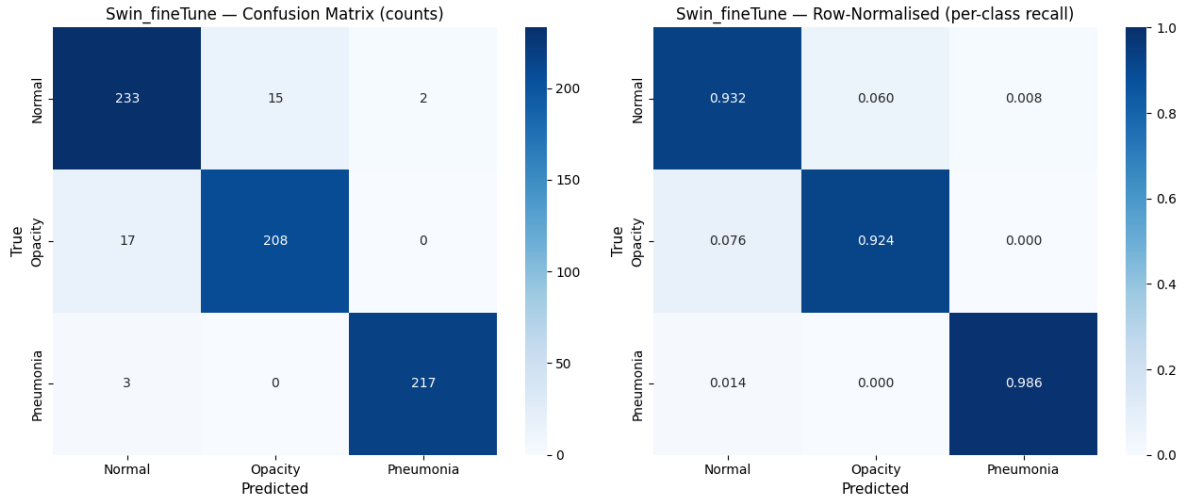


Figure 3: Confusion matrices: raw counts (left) and row-normalised per-class recall (right).

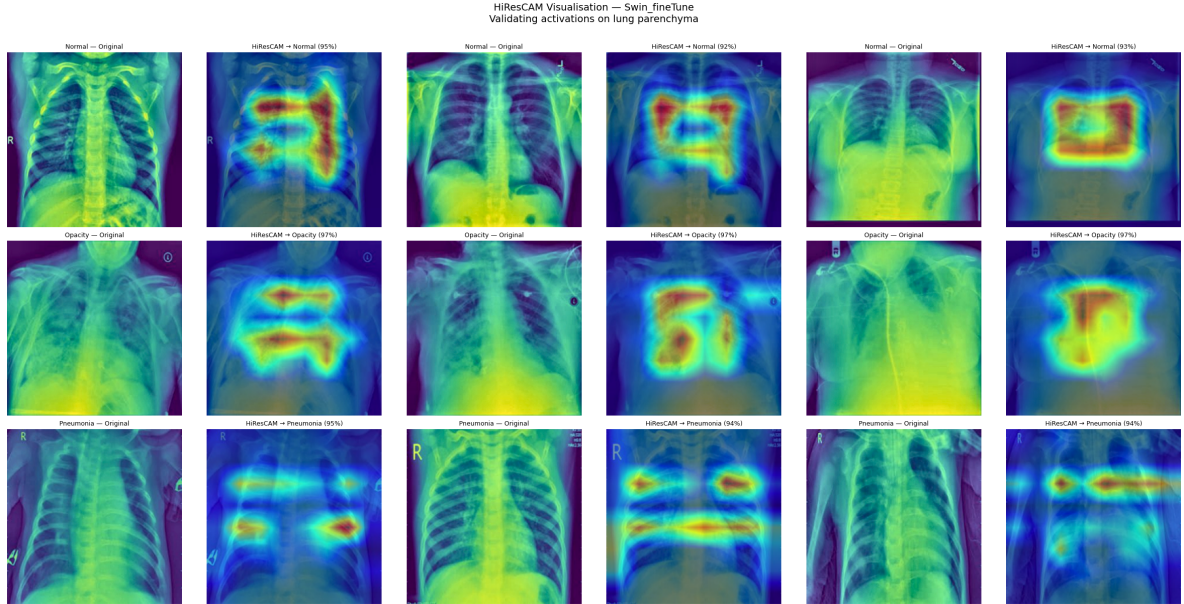


Figure 4: HiResCAM heatmaps (Swin V2-S fine-tune). Activations localise to lung fields, validating clinically correct learned representations.

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